DYUTIDEEPTA BANERJEE

Physical AI-Machine Vision Specialist

+91-9845210699 | dyutideepta.banerjee@gmail.com | Bangalore, India, 560077
https://github.com/DyutideeptaB | https://www.linkedin.com/in/dyutideepta-banerjee/

SUMMARY

AI/ML/Vision professional with 4+ years of cross-domain experience in Physics-based modelling and production-grade computer vision systems. Proven track record using Python and MATLAB, together with modern AI tooling, to build efficient, scalable solutions, automate workflows, and drive industrial innovation across remote sensing, robotics, spacetech, defence, and digital twins. Motivated to tackle complex, real-world problems within high-impact, innovative teams.

EXPERIENCE

Data Scientist
Spatialty.AI
09/2025 - 01/2026 | Bangalore, India

- Built end-to-end Python pipelines that combined Physics-informed modelling and Computer Vision, enabling high-fidelity simulations, automated synthetic data generation and feature analytics for defence and construction projects.
- Delivered robust constraint-based algorithms, improving accuracy and reliability of change monitoring across 2D, 3D, thermal & magnetic image streams in the record fixed-term collaboration.

Cross-Platform Solutions Expert
Johnson & Johnson
06/2024 - 08/2025 | Bangalore, India

- Consulted for J&J global stakeholders to define business solutions, conduct cross-platform quality monitoring & testing, and deliver actionable improvements for the product lookup system.
- Managed sprint cycles with international teams across time zones, streamlining agile processes to reduce fix turnaround by 40%.

Software Automation Specialist
Toriox PRJ Packaging Pvt Ltd
09/2024 - 12/2024 | Bangalore, India

- Designed and deployed edge-based Python modules for real-time inventory management, reducing manual intervention by 80% in low-latency production environments. This was a freelance work.

Artificial Intelligence Research Intern *Project Link*
MathWorks: Customer Success Engineering Group
11/2022 - 05/2023 | Munich, Germany

- Established a standardized Computer Vision pipeline in MATLAB for efficient large-scale multispectral remote sensing data analysis.
- Developed a specialized MATLAB AI application to automate spectral image preprocessing, utilizing techniques like segmentation, enhancement & annotation to generate high-quality datasets.
- Engineered and adapted neural network models for machine and deep learning, consistently achieving over 95% prediction accuracy across diverse cross-domain applications.

Visiting Fellow
International Center for Theoretical Sciences Tata Institute of Fundamental Research
07/2019 - 04/2020 | Bangalore, India

- Developed and validated Physics-based simulation models for complex turbulent flows, using quantitative motion analysis techniques.
- Performed detailed motion analysis by utilizing videographic data analysis to rigorously assess the accuracy of drag models.
- Designed and created custom experimental setups, including the use of 3D printing, to generate data for subsequent analysis and model validation.

EDUCATION

MSc Physics of Data
University of Padua
10/2020 - 04/2024 | Padua, Italy
(2022/23 foreign exchange)

BSc (Hons) Physics, minor in Data Science
Azim Premji University
08/2016 - 08/2019 | Bangalore, India

SKILLS

Computational Physics & Modelling
Strong foundation in Physical sciences with expertise in AI for simulations, algorithm development, automation & Physics-based modelling

AI & Data Science
- Machine & Deep Learning Frameworks, Predictive Analytics
- Data Wrangling-Processing-Interpretation, Statistical Analysis, Data Modelling
- Computer Vision, Point cloud, Spectral Data Analysis
- Technical Documentation, Version Control and Quality Assurance

Programming & Tools
Python (libraries: NumPy, Pandas, Matplotlib, TensorFlow, Keras, PyTorch, Scikit, Open3D, PyBullet, Tkinter, Fernet), MATLAB, R, TeX
CAD, GIS, MS Office
Git, Bash, Jira

Soft Skills
Excellent communication, public speaking, presentation and team leadership skills, financial and business acumen

CERTIFICATIONS

Finance for Non-finance Professionals 2025
Rice University by Coursera *Certificate*

Deep Learning Onramp 2023
MathWorks *Certificate*

Introduction to Cosmology 2017
M. P. Birla Institute of Fundamental Research

Programming with Python 2017
University of Michigan by Coursera *Certificate*

R for Data Science 2016
Miscrosoft by edX *Certificate*

SELECT TECHNICAL CONTRIBUTIONS

Visual-Guided Dual-Arm Synchronisation with Decentralised Control

Designed, implemented & simulated a system for two robotic arms to achieve motion using visual feedback with Python PyBullet codes. Utilising input from real-time tracking and pose regulation by the second robotic arm, optimized object transfer efficiency in simulated environments.

Encrypted QR generator with Decryption Algorithm *Project Link*

Built an open source project, a Python-based toolkit for encrypted QR code generation with data created with Tkinter GUI, enhancing data security with Fernet-based decryption system.

Built an AI-Powered Ed-tech Platform for NASA Space Apps Hackathon 2024 *View Platform*

Created automated data pipelines for acquiring NASA exoplanetary datasets
Deployed NLP/LLM modules like Llama for improved data processing and tailored course material based on user feedback

Studied Bristle Bots as a minimal model of directed motion

Designed & developed 3D printed miniaturised bots having bristle-like legs for tailoring directional movement, with an electromotive light-sensing circuit. Conducted experimental analysis and validation for the bot's motion exposed to an environmental gradient
Formalised theory for design adept towards directed motion to initialise study for collective dynamic systems

INVITED SPEAKER

Seminar Speaker for the Applicability of AI in Planetary Feature Detections
Max Planck Institute for Solar System Research
05/2023 Göttingen, Germany

Scientific Exhibitor for Buffon's Mathematical Model
Indian Institute of Science
02/2020 Bangalore, India

VOLUNTEERING & LEADERSHIP

Academic Coach for graduate-level Physics & Mathematics
Classgap by GoStudent
01/2024 - 12/2024 Global

Communication Coach
Center for Linguistics, University of Padua
12/2021 - 12/2022 Padua, Italy

Certified Peer Counsellor for Mental Wellness
Azim Premji University
07/2017 - 04/2019 Bangalore, India

Special-ed Teacher Volunteer for the underprivileged at an NGO
Parikrma Humanity Foundation
07/2017 - 12/2017 Bangalore, India

KEY ACHIEVEMENTS

First-authored Manuscript under review

Focused on modular vision pipelines for multi-dataset analysis in Lunar data. Co-authored by Max Planck Institute, University of Padua & MathWorks.

Winner of ERODEM Research Grant 2024

Italian Ministry's funding for conservation of heritage sites by carbon dating using ML models

Poster Publication at 11th European Lunar Symposium 2023 *View Poster*

AI-based detection of Lunar Sinuous rilles: A comparison with manual detection methods

Winner of Erasmus + Traineeship Grant 2022

LANGUAGES

English
Native

Hindi
Native

Bengali
Mother Tongue

Italian
Beginner

German
Beginner